

Power Budget - OLED

Team Number:	Team 307
Project Name:	Submersible Exploration Device
Team Member Names:	Abriana Poola

A. List ALL major components (active devices, integrated circuits, etc.) except for power sources, voltage regulators,

All Major Components	Component Name	Part Number	Voltage	Current (mA)	Unit
	ESP32-S3-WROOM-1	ESP32-S3-WR	3.0V - 3.6V	1	500 mA
	OLED display module	B085WCRS7C	3.3V	1	25 mA
	LED 1	Generic LED	3.3V	1	6 mA
					6 mA

B. Assign each major component above to ONE power rail below. Try to minimize the number of different power rails in

+12V Power Rail	Component Name	Part Number	Voltage(V)	Current(mA)	Unit
				0	mA
				0	mA
				0	mA
				0	mA
				0	mA
				0	mA
				Subtotal	
				Safety Margin	25%
				Total Current Required on +12V Rail	0 mA
c1. Regulator or Source Ch1	Direct +12V input from wa Barrel jack J3	12V nominal	1	1000	1000 mA
				Total Remaining Current Available on +12V Rail	1000 mA

+3.3V Power Rail	Component Name	Part Number	Voltage (V)	Current (mA)	Unit
------------------	----------------	-------------	-------------	--------------	------

	ESP32-S3-WROOM-1	ESP32-S3-WR	3.0V - 3.6V	1	500	500	mA
	OLED display module	B085WCRS7C	3.3V	1	25	25	mA
	LED 1	Generic LED	3.3V	1	6	6	mA
						0	mA
						531	mA
						25%	
						Total Current Required on +3.3V Rail	663.75 mA
c2. Regulator or Source Ch	+3.3V buck regulator	LM2576HVS-34V - 60V input		1	3000	3000	mA
						Total Remaining Current Available on 3.3V Rail	2336.25 mA

D. Select a specific external power source (wall supply or battery) for your system, and confirm that it can supply all of

External Power Source 1	Component Name	Part Number	Voltage (V)	Rated Output Current (mA)	Maximum Current (mA)	Current (mA)	Unit
Power Source 1 Selection	12V wall adapter	B0D269PYK9	AC input	12V	1000	1000	mA
Power Rails Connected to External Power Source 1	3.3 Regulator	LM2575D2T-3.3	5V 12V source	1	228	228	mA
						0	mA
						0	mA
Total Remaining Current Available on External Power Source 1						772	mA
External Power Source 2	Component Name	Part Number	Voltage (V)	Rated Output Current (mA)	Maximum Current (mA)	Current (mA)	Unit
Power Source 2 Selection	Not used	N/A	N/A	N/A	0	0	mA
Power Rails Connected to External Power Source 2						0	mA
						0	mA
Total Remaining Current Available on External Power Source 2						0	mA
Battery Life							hours

Notes / Assumptions

ESP32 budget uses 500 mA peak design current so the 3.3 V rail can handle Wi-Fi startup/current spikes. OLED current estimated at 25 mA because the schematic does not include the OLED module part number. LED current assumes 220Ω series resistors and ~2.0V LED forward drop: $(3.3V - 2.0V)/220\Omega \approx 5.9mA$ each. External 12V source current is estimated from 3.3V rail load using 80% buck efficiency.